



Applied Chemistry A

T2-172

Friday, 11 July 2025

02:00 PM -05:00 PM

Names:

Index number

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

QUESTIONS and ANSWERS BOOKLET

TRADE (s): MAT, AUT, PLT, WOT, FOR, AGR, ANH, BDC, FPA,FOP, IND, LSV,
LET, PWO, WIR

SUBJECT: APPLIED CHEMISTRY A

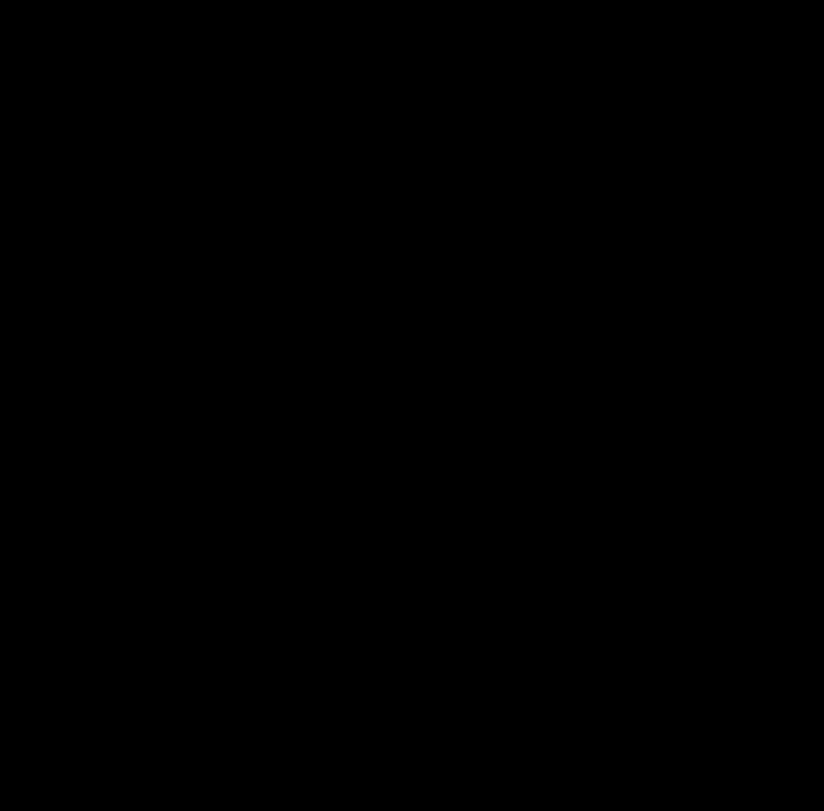
DURATION: 3 Hours

Read carefully the instructions on page 3

TOTAL MARKS...../100

FOR EXAMINER'S USE ONLY

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INSTRUCTION TO CANDIDATE:

1. This Exam paper is composed of **three sections (A, B and C)**. Follow the instructions given below and answer the indicated questions for a total of **100 marks**.

Section **A**: All questions are **Compulsory** **55 marks**

Section **B**: Attempt any three (3) **30 marks**

Section **C**: Attempt anyone (1) **15 marks**

2. Fill in your actual **names** and **Index number** on the provided space(Cover).
3. Do not remove or tear any page or part of it of this Questions and Answers Booklet.
4. Answer in the language in which the Examination is set.
5. In section **A** answer questions in the provided space. In case of multiple choice questions, just **circle** the letter corresponding to the correct option.
6. In sections **B** and **C**, answer questions in the provided space after section C questions.
7. **You do not need the Periodic Table.**
8. **Allowed materials:** - Blue or Black pen.
- Geometrical Instruments
- Silent non-programmable calculators

SECTION A: Attempt all questions**(55 marks)**

01. Fill in the blank spaces, of the following sentences, with the provided underlined terms. Each term will be used once:

(sub-atomic, neutrons, neutral).

- i) An atom is electrically because of equal number of positively charged protons and negatively charged electrons. **(1mark)**
- ii) Protons and.....are found inside the nucleus. **(1mark)**
- iii) There are three main..... particles in the atom. **(1mark)**

02. Circle the letter corresponding to the correct answer for each of the following sub-questions.

- i) What do burette used for in laboratory setting? **(1Mark)**
 - A) To measure and transfer large volumes accurately.
 - B) To heat substances to high temperature.
 - C) To measure and dispense precise volume of liquid during titration.
 - D) To mix chemical solutions thoroughly.
- ii) What do acidimetry mean among the following descriptions? **(1Mark)**
 - A. A technique for determining the concentration of an acid in a solution.
 - B. A method used for to measure the basicity of a solution.
 - C. The process of neutralizing an acid a base with a base.
 - D. A method for measuring the pH level of a solution.
- iii) Which of the following options that correctly defines the term alkalimetry? **(1Mark)**
 - A. A method used for to measure the acidity of a solution.
 - B. A technique used for determining the concentration of a base in a solution.
 - C. The process of neutralizing an acid a base with a base.
 - D. A method for measuring the pH level of a solution.

03. Using True(**T**) or False (**F**) state whether each of the following statements about organic compounds are correct or wrong.

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- i) The general molecular formula for alkanes is C_nH_{2n+2} **(1Mark)**
- ii) Organic compounds generally have higher melting points than inorganic compounds. **(1Mark)**
- iii) Organic compounds are typically less volatile than inorganic compounds. **(1Mark)**

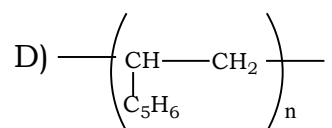
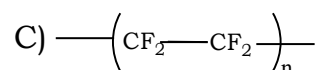
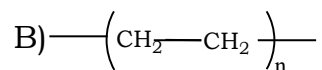
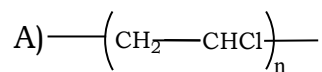
- 04.** Match using middle column each class of carbohydrates in the left column, with its corresponding example, in the right column of the table. **(3Marks)**

Class of carbohydrates	Answer	Example
1. Monosaccharides	1=	A) Sucrose
2. Disaccharides	2=	B) Starch
3. Polysaccharides	3=	C) Glucose

- 05.** Circle the letter corresponding to the correct answer to each of the following questions. **(1Mark)**
- i) What is a monomer? **(1Mark)**
- A) A large molecule made up of many repeating subunits.
- B) A single, small molecule that can combine with others to form a polymer.
- C) A type of lipid used in cell membrane.
- D) A complex protein that catalyzes reactions in living organisms.
- ii) Which of the following polymers is a synthetic condensation polymer? **(1Mark)**
- A) Polyethene
- B) Polystyrene
- C) Polyester
- D) Polypropene

iii) Which of the following repeating units is for polyvinylchloride (PVC)?

(1Mark)



06. Fill the blank spaces in the following sentences, with appropriate terms provided as: (**distillation, spectroscopy, gravimetry**).

i) is a process which, in qualitative chemical analysis consists separating the components of a mixture based on difference in their boiling points.

(1Mark)

ii) The is a method of quantitative chemical analysis where the amount of a substance is determined by measuring the mass of a solid product separated from the sample.

(1Mark)

iii) In qualitative chemical analysis,is the study of the interaction between matter and electromagnetic radiation as a function of wavelength or frequency of the radiation.

(1Mark)

07. Circle the letter corresponding to the correct answer to each of the following sub-questions.

i) Identify among the following list, two correct factors that affect the solubility of salts.

(2Marks)

A) Color of the salt

B) Molecular size of the salt

C) Shape of the solvent

D) Polarity of the salt

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- ii) Identify two salt which are **NOT** soluble in water, among the following list.

(2Marks)

- A) KCl
- B) PbCl₂
- C) CaSO₄
- D) (NH₄)₂SO₄

08. Circle the letter corresponding to the correct answer to each of the following sub-questions.

- i) Which two (2) elements among the following are used to find the position of an element on the periodic table?

(2Marks)

- A) Atomic number
- B) Atomic mass
- C) Electron configuration
- D) Number of neutrons

- ii) Why is an atom said to be electrically neutral?

(2Marks)

- A) An atom has more protons than electrons.
- B) The number of protons is equal to the number of electrons and their charges cancel each other.
- C) The number of neutrons is equal to the number of electrons and their charges cancel each other.

09. Use the column in the middle with number given to chemical equation and match it to the roman number given to the type of reaction in the table below.

(4Marks)

Chemical reaction	Answer	Type of reaction
1. $\text{C(s)} + \text{O}_2\text{(g)} \longrightarrow \text{CO}_2\text{(g)}$	1.=....	i. Decomposition reaction
2. $\text{HCl(aq)} + \text{KOH(aq)} \longrightarrow \text{KCl(aq)} + \text{H}_2\text{O(aq)}$	2.=....	ii. Combustion reaction
3. $\text{CaCO}_3\text{(aq)} \longrightarrow \text{CaO(s)} + \text{CO}_2\text{(g)}$	3.=....	iii. Precipitation reaction
4. $\text{AgNO}_3\text{(aq)} + \text{NaCl(aq)} \longrightarrow \text{AgCl(s)} + \text{NaNO}_3\text{(aq)}$	4.=....	iv. Neutralization reaction

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10. Using **True(T)** or **False (F)** state whether each of the following statements about organic compounds are correct or wrong.

- i) Silicon carbide (SiC) is an example of organic compound
..... **(1Mark)**
- ii) Alicyclic organic compounds are compounds with open carbon chains only. **(1Mark)**
- iii) Generally, organic compounds are insoluble in water. **(1Mark)**
- iv) The compound, $\text{CH}_3\text{CH}_2\text{CHNH}_2$ is an example of hydrocarbon compound. **(1Mark)**

11. Fill the blank spaces in the following sentences, with appropriate terms provided as: (**insulin, nucleotides, amino acids, ribonucleic acid**).

- i) The nucleic acids are macromolecules composed of **(1Mark)**
.....
- ii) An example of nucleic acid is **(1Mark)**
- iii) Proteins are large macromolecules that are made by condensation of **(1Mark)**
- iv) An example of protein is..... **(1Mark)**

12. Circle the letter corresponding to the correct answer for each of the following sub-questions.

- i)** A condensation polymer is a polymer: **(1Mark)**
- A) formed by the repeated joining between monomers possessing single bonds only.
- B) formed by the repeated joining of two different monofunctional monomer units.
- C) formed by the repeated joining of two different bifunctional monomer units.
- D) formed by the repeated joining between monomers possessing double bonds only.

ii) An addition polymer is: **(1Mark)**

- A) formed by the repeated joining of unsaturated monomers (each monomer contains at least one double bond).
- B) formed by the repeated joining of monomers possessing single bonds only.
- C) formed by the repeated joining of two different bifunctional monomer units.
- D) formed by the repeated joining between one saturated monomer and one unsaturated monomer.

iii) Which of the following is a natural polymer? **(1Mark)**

- A) Cellulose
- B) Polypropene
- C) Polystyrene
- D) Polyvinylchloride

iv) Which of the following polymers is a synthetic polyamide? **(1Mark)**

- A) Terylene
- B) Nylon-6,6
- C) Cellulose
- D) Deoxyribonucleic acid

13. Circle the letter corresponding to the correct answer for each of the following sub-questions. An atom of element **Y** has the following representation ${}^{37}_{17}\text{Y}$.

i) How many protons does **Y** have? **(1Mark)**

- A) 17 protons
- B) 20 protons
- C) 37 protons
- D) 54 protons

ii) How many electrons does **Y** have? **(1Mark)**

- A) 54 electrons
- B) 37 electrons
- C) 20 electrons
- D) 17 electrons

iii) How many neutrons does **Y** have? **(1Mark)**

- A) 54 neutrons
- B) 37 neutrons
- C) 20 neutrons
- D) 17 neutrons

iv) What is the correct electron configuration for ${}^{37}_{17}\text{Y}$? **(1Mark)**

- A) $1s^2 2s^2 3p^6 3s^2 3p^6 4s^2$
- B) $1s^2 2s^2 3p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1$
- C) $1s^2 2s^2 3p^6 3s^2 3p^5$
- D) $1s^2 2s^2 3p^6 3s^2 3p^6$

14. Circle the letter corresponding to the correct answer for each of the following sub-questions.

i) Which two (2) of the following substances give neutral solutions, with pH=7? **(2Marks)**

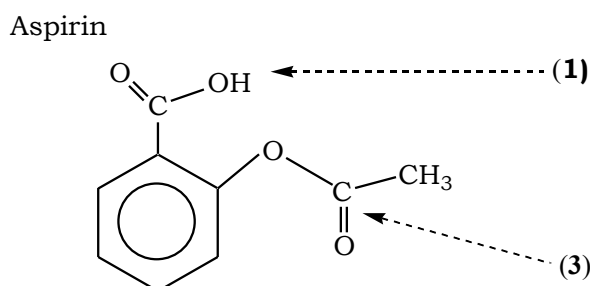
- A) Nitric acid (HNO_3)
- B) Sodium chloride (NaCl)
- C) Potassium hydroxide (KOH)
- D) Sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)

ii) What is the molarity of a solution made by adding 11 g of sodium chloride (NaCl) to make 250 ml of solution? **(2Marks)**

- A) 0.188mol/l
- B) 0.0044 mol/l
- C) 44 mol/l
- D) 0.752 mol/l

(Atomic masses: Na =23, Cl =35.5)

- 15.** The structure of aspirin, a pharmaceutical compound used as pain reliever and anti-inflammatory, structure is shown below. **Circle the letter corresponding to the correct answer for each of the following sub-questions**



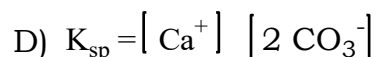
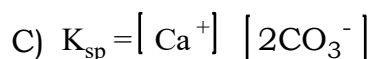
- i) To which class of organic compounds does aspirin belong? **(2Marks)**
- A) Hydrocarbon compound
 B) Aliphatic compound
 C) Aromatic compound
 D) Alicyclic compound
- ii) Name the three functional groups found in aspirin.
- a) (1) = **(1Mark)**
 b) (2) = **(1Mark)**
 c) (3) = **(1Mark)**

Section B: Attempt any three (3) questions

(30 marks)

- 16.** Given the two slightly soluble electrolytes with their respective K_{sp} values.
- $\text{CaCO}_3 \quad K_{sp} = 2.8 \times 10^{-9}$
 $\text{Ag}_2\text{CO}_3 \quad K_{sp} = 8.1 \times 10^{-12}$

- i) Which among the following expressions, is correct for the solubility product constant, K_{sp} , for CaCO_3 ? **(2Marks)**
- A) $K_{sp} = [2\text{Ca}^+] [\text{CO}_3^{2-}]$
 B) $K_{sp} = [\text{Ca}^{2+}] [\text{CO}_3^{2-}]$



ii) Choose the right mathematical value of solubility S for Ag_2CO_3 . **(4Marks)**

A) 1.265×10^{-4}

B) 1.423×10^{-6}

C) 2.0×10^{-4}

D) 2.0×10^{-12}

iii) Select the correct unit, in mol/dm^3 of the solubility product constant, K_{sp} , for $CaCO_3$? **(2Marks)**

A) mol^3/dm^9

B) mol^2/dm^6

C) dm^6/mol^2

D) mol^2/dm^9

iv) Which among the following is the correct unit, in mol/dm^3 of the solubility product constant, K_{sp} , for Ag_2CO_3 ? **(2Marks)**

A) dm^9/mol^3

B) mol^2/dm^6

C) dm^6/mol^2

D) mol^3/dm^9

17. Standard solutions are prepared either by dissolving a solid chemical or by diluting the liquid stock solution.

$$[\text{Molarity} = \frac{d \times \% \times 10}{Mm}] \text{ and } [M_1V_1 = M_2V_2]$$

Where d = density, % = percentage of purity of substance,

Mm= molar mass M_1 , M_2 , V_1 and V_2 are the molarity and volume

of solution before and after dilution. *Circle the letter corresponding to the correct answer for each of the following sub-questions*

i) Which of the following solutions whose concentration is determined in a titration? **(1Mark)**

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- A) Titrant
B) Indicator
C) Analyte
D) Solvent
- ii) Which of the following, is a best apparatus specifically used in titration? **(1mark)**
A) Centrifuge
B) Burette
C) Bunsen Burner
D) Thermometer
- iii) A bottle of stock solution of sulphuric acid (H_2SO_4) is labeled with $d = 1.84$, $\% = 98$, and $M_m = 98\text{g/mol}$. Determine the concentration in mol/l, of this stock solution. **(3Marks)**
A) 16.4 mol/l
B) 18.1 mol/l
C) 16.4 mol/l
D) 18.4 mol/l
- iv) What volume of a 12 mol/dm^3 HCl required to prepare 500 cm^3 of a 0.5 mol/l HCl solution? **(3Marks)**
A) 20.83 cm^3 of HCl
B) 10.83 cm^3 of HCl
C) 41.66 cm^3 of HCl
D) 83.20 cm^3 of HCl
- v) Which two (2) of the following are malpractices that can be source of errors when preparing solutions in laboratory? **(2Marks)**
A) Make proper calculations methods.
B) Use glassware which is not properly cleaned.
C) Incorrect reading of the mass weighed on the balance.
D) Understand well the solution requirements.
(Consider atomic masses: $\text{H}=1$, $\text{O}=16$, $\text{S}=32$, $\text{Cl}=35.5$)

18. The chemical analysis may be qualitative or quantitative.

i) Identify two factors among the following considered when conducting the qualitative analysis. **(2Marks)**

- A) Absorption of light
- B) Density
- C) Odor
- D) Solubility

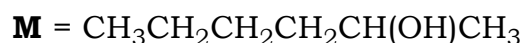
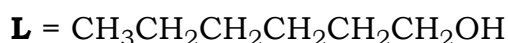
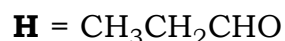
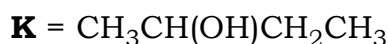
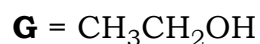
ii) Select two (2) quantitative methods among the following. **(2Marks)**

- A) Addition
- B) Titration
- C) Gravimetry
- D) Flame test

iii) Complete the following table to show if each of the following processes is of qualitative analysis (**QLI**) or quantitative analysis (**QNT**). **(6Marks)**

Process	QLI or QNT
Determination of the unknown molarity using the standard solution of NaOH.	
Titrate the solution Y with the solution of HCl only with the purpose of confirming if X is a base or NOT.	
Identification of potassium ions (K^+) based on its characteristic color.	
Detection of carbon dioxide (CO_2) by bubbling the gas through limewater.	
Determination of the mass of $BaSO_4$ through precipitation reaction.	
Identification of the type of functional group present in an organic compound.	

19. Given a list of six organic compounds below.



i) Among the following options which one that correctly define “Homologous series”?

(2Marks)

- A) A series of compounds that have no common functional group but have similar physical properties.
- B) A series of compounds with similar chemical properties and some functional group differing from the successive member by $-\text{CH}_2-$ group.
- C) A series of compounds that that share the same functional group and differ by a single carbon atom in their structure.
- D) A series of compounds with identical molecular formulae but different structural arrangements.

ii) Give the IUPAC names for the compounds **G**, **H**, **J**, and **K**.

(4Marks)

iii) Identify two pairs of compounds, from the list above, that belong to the same homologous series.

(4Marks)

20. The table below shows the standard reduction potentials for two substances.

Electrode reaction	$E^\ominus / \text{V}$
$\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \rightleftharpoons \text{Cr}(\text{s})$	-0.74
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Fe}(\text{s})$	-0.44

A complete electrochemical cell $\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \rightleftharpoons \text{Cr}(\text{s})$ is obtained when the half-cell of chromium is connected to the half-cell of iron with all required accessories.

i) Which of the two electrodes will be the:

a) anode?

(1Mark)

b) cathode?

(1Mark)

ii) Write the equation that takes place at the:

- a) anode. (2Marks)
- b) cathode. (2Marks)
- iii) Calculate the cell potential for the above electrochemical cell. (2Marks)
- iv) Which of the two substances is the reducing agent? Justify your answer. (1marks)

Section C: Attempt only one (1) question**(15 marks)****21.** This question concerns inorganic substances.

- i) Match each oxide on the left, with its corresponding acid-base character on the right, in the table below.

(4Marks)

Oxide	Answer	Acid-base character
A) Na ₂ O	A) =	1. Acidic oxide
B) SO ₂	B) =	2. Basic oxide
C) Al ₂ O ₃	C) =	3. Amphoteric oxide
D) CO	D) =	4. Neutral oxide

- ii) Use **True (T)** or **False (F)** to state whether each of the following statements about chemical substances are right or wrong.

a) The pH of a 0.1 M HCl solution is equal to 13. (2Marks)

b) The molarity of a 100 ml solution containing 1.17 g of NaCl is equal to 0.2 mol/l. (2Marks)

c) The concentration in hydrogen ions, [H⁺], for a HNO₃ solution with pH = 2, is equal to 0.001 mol/l. (2Marks)

d) The solution of KCl with pH = 3 has no effect neither on the red litmus paper nor the blue litmus paper. (2Marks)

(Atomic masses: H =1, N =14, O=16, Na = 23, Cl =35.5)

- iii) Use the substances, Mg, O₂, and H₂O as the only starting materials to show how you can prepare magnesium hydroxide, Mg(OH)₂(aq) through one or more steps. (3Marks)

22. This question is about electrolytic and electrochemical cells.

i) Match each type of half-cell in the left column, with its corresponding name in the right column of the table below.

(3Marks)

Half-cell type	Answer	Half-cell name
A) Ag^+/Ag	A) =	1. Ion half cell
B) H^+/H_2	B) =	2. Metal-metal ion half cell
C) $\text{Fe}^{3+}, \text{Fe}^{2+}/\text{Pt}$	C) =	3. Non-metal half cell

ii) Match each term in the left column, with its corresponding meaning in the right column of the table below.

(5Marks)

Term	Answer	Meaning
A) Reduction reaction	A) = ...	1. Device used in electrochemical cells to maintain electrical neutrality by allowing the flow of ion between the two half-cells.
B) Anode	B) = ...	2. Reaction by which an atom, ion, or molecule loses electrons.
C) Oxidation reaction	C) = ...	3. Electrode where oxidation reaction occurs.
D) Cathode	D) = ...	4. Electrode where reduction reaction occurs.
E) Salt bridge	E) = ...	5. Reaction by which an atom, ion, or molecule gains electrons.

iii) Construct an electrolytic cell which shows the electrolysis of molten sodium chloride, NaCl(l) . The diagram should be well labelled showing the positive electrode, the negative electrode, the battery with its polarities, electrolyte, the direction of electrons in the external circuit, as well as the equations that take place at both electrodes.

(7Marks)

END

RESERVED SPACE FOR SECTION B AND C ANSWERS

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